Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Data Analytics with R

Course Code: C182

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	Initial release for C182: two-day, 15-hour course (9:30am – 5:30pm) with five topics, ten R labs and Excel evidence workbooks.	Dr Alfred Ang

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TABLE OF CONTENTS

Course Overview **4**

Learning Outcomes **4**

Daily Schedule **4**

Day 1 (9:30am – 5:30pm): Prepare, summarise and analyse quantitative research data 4

Day 2 (9:30am – 5:30pm): Analyse text, visualise evidence and interpret findings 4

Topic-by-Topic Delivery Guide **5**

Topic 01: Data Preparation and Transformation 5

Topic 02: Data Summary 6

Topic 03: Quantitative Data Analysis 6

Topic 04: Qualitative Data Analysis 6

Topic 05: Data Visualization 7

Resources Required 7

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Course Overview

Data Analytics with R is a two-day, 15-hour short course (9:30am – 5:30pm each day). Learners progress from controlled preparation through quantitative and qualitative analysis to reproducible visual communication in R.

Course Code	C182
Training Duration	15 hours across 2 days (9:30am – 5:30pm)
Trainer	Dr Alfred Ang
Mode	Instructor-led classroom with individual R labs and Excel evidence workbooks

Learning Outcomes

- LO1: Prepare and transform data for analysis using R.
- LO2: Summarise data with descriptive statistics, figures, graphs and tables.
- LO3: Perform quantitative data analysis and interpret the findings.
- LO4: Perform qualitative data analysis and determine significance or effect.
- LO5: Create effective data visualisations in R and interpret research findings.

Daily Schedule

Day 1 (9:30am – 5:30pm): Prepare, summarise and analyse quantitative research data

Time	Duration	Topic / Activity	Method	Slides
9:30-9:50	20 min	Welcome, introductions, outcomes and evidence conventions	Facilitated briefing	—
9:50-10:50	60 min	Topic 1: research-data workflow, tidy structure, import, missingness, transformation, joins and reconciliation	Concept briefing, editable chart examples and questioning	Slides 10-34
10:50-12:40	110 min	Labs 1-2: prepare dengue data; join, reshape and aggregate vaccination data	Demonstration, individual R practice and evidence debrief	Slides 30-33
12:40-13:10	30 min	Lunch break	Lunch	—
13:10-14:00	50 min	Topic 2: data types, centre, spread, grouped summaries, distributions and sampling	Concept briefing, editable chart examples and questioning	Slides 35-59
14:00-15:00	60 min	Labs 3-4: descriptive statistics, grouped tables and basic plots	Demonstration, individual R practice and evidence debrief	Slides 55-58
15:00-15:50	50 min	Topic 3: correlation, regression, hypothesis tests, effect size and ANOVA	Concept briefing, editable chart examples and questioning	Slides 60-84
15:50-17:10	80 min	Labs 5-6: correlation/regression and tests/group comparison	Demonstration, individual R practice and evidence debrief	Slides 80-83
17:10-17:30	20 min	Day 1 evidence review and Q&A	Reflection and Q&A	—

Instructional time excluding the 30-minute lunch: 450 minutes (7.5 hours, 9:30am – 5:30pm).

Day 2 (9:30am – 5:30pm): Analyse text, visualise evidence and interpret findings

Time	Duration	Topic / Activity	Method	Slides
9:30-9:50	20 min	Day 1 recap	Facilitated briefing	—

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9:50-11:10	80 min	Topic 4: qualitative evidence, text preparation, tokenisation, stop words, frequency and coding	Concept briefing, editable chart examples and questioning	Slides 86-110
11:10-12:20	70 min	Lab 7: prepare text and analyse term frequency	Demonstration, individual R practice and evidence debrief	Slides 106-107
12:20-12:50	30 min	Lunch break	Lunch	—
12:50-13:55	65 min	Lab 8: create and critique a reproducible word cloud	Demonstration, individual R practice and evidence debrief	Slides 108-109
13:55-14:20	25 min	Topic 5 introduction: grammar of graphics and question-to-chart fit	Concept briefing, editable chart examples and questioning	Slides 111-135
14:20-15:35	75 min	Topic 5: scales, geoms, facets, accessibility, visual QA and interpretation	Concept briefing, editable chart examples and questioning	Slides 111-135
15:35-16:55	80 min	Labs 9-10: layered visuals and capstone evidence briefing	Demonstration, individual R practice and evidence debrief	Slides 131-134
16:55-17:30	35 min	Course recap, Q&A, course feedback and next steps	Facilitated review and Q&A	Slides 136-139

Instructional time excluding the 30-minute lunch: 450 minutes (7.5 hours, 9:30am – 5:30pm).

Topic-by-Topic Delivery Guide

Topic 01: Data Preparation and Transformation · Slides 10-34

Research-data workflow · import/export · tidy structure · cleaning · joins · transformation

- **Question before code:** Define the decision, population, variables and unit of observation before transforming data.
- **Tidy structure:** Each variable is a column, each observation is a row and each value occupies one cell.
- **Typed imports:** Dates, factors, text and numbers need deliberate types so analysis behaves predictably.
- **Transformation pipeline:** Select, filter, mutate, join, reshape and aggregate as visible reproducible stages.
- **Missingness is information:** Distinguish structurally absent, unknown and invalid values before removal or imputation.
- **Reconciliation:** Prove row counts, keys and material totals before and after every transformation.

Lab 1: Prepare and Transform Dengue Surveillance Data · Slides 30-31

Alignment: LO1. R script: lab-01-data-preparation.R. Excel evidence: lab-01-data-preparation.xlsx. Learner output: A reproducible clean dengue time-series dataset, transformation log and reconciliation evidence. Acceptance: The script runs without error, output rows reconcile to input less documented exclusions, dates are valid, and the clean CSV and plot are created.

Lab 2: Join, Reshape and Aggregate Vaccination Data · Slides 32-33

Alignment: LO1. R script: lab-02-join-reshape.R. Excel evidence: lab-02-join-reshape.xlsx. Learner output: A joined long-form vaccination dataset, unmatched-key report, annual summary and transformation audit. Acceptance: The left join preserves all source rows, unmatched keys are zero or documented, and total doses reconcile across input, long and wide summaries.

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Topic 02: Data Summary · Slides 35-59

Data types · descriptive statistics · distribution · grouped summaries · basic plots and tables

- **Variable type drives method:** Nominal, ordinal, discrete and continuous data support different summaries and charts.
- **Centre:** Mean and median answer different questions when the distribution is skewed or contains outliers.
- **Spread:** Range, interquartile range, variance and standard deviation quantify variability.
- **Grouped summaries:** Compare meaningful sub-populations while preserving denominators and sample sizes.
- **Sampling variability:** A statistic changes from sample to sample; uncertainty is part of the result.
- **Review accuracy:** Recompute, reconcile and inspect distributions before accepting a summary.

Lab 3: Summarise Centre, Spread and Distribution · Slides 55-56

Alignment: LO2. R script: lab-03-descriptive-statistics.R. Excel evidence: lab-03-descriptive-statistics.xlsx. Learner output: A descriptive-statistics table, distribution plot, outlier review and accuracy checklist. Acceptance: R and Excel checks agree, units and sample size are stated, outliers remain traceable, and the chosen centre measure matches the observed distribution.

Lab 4: Build Grouped Tables and Basic Plots · Slides 57-58

Alignment: LO2. R script: lab-04-grouped-summary.R. Excel evidence: lab-04-grouped-summary.xlsx. Learner output: Frequency/proportion tables, grouped numerical summaries, bar and box plots, and a denominator audit. Acceptance: Counts reconcile to the dataset, proportions sum to 100% within rounding, every grouped statistic includes n, and the written comparison matches both tables and plots.

Topic 03: Quantitative Data Analysis · Slides 60-84

Correlation · regression · hypothesis testing · effect size · ANOVA · interpretation

- **Correlation:** Measures strength and direction of association; it does not by itself establish causation.
- **Regression:** Models an outcome as a function of one or more predictors under explicit assumptions.
- **Hypothesis testing:** Compares observed evidence with a pre-specified null model and significance threshold.
- **Effect size:** Quantifies how large the difference or relationship is in practical terms.
- **ANOVA:** Tests whether group means differ beyond what within-group variability would suggest.
- **Interpretation boundary:** Separate evidence, assumptions, uncertainty and business or research judgement.

Lab 5: Analyse Correlation and Regression · Slides 80-81

Alignment: LO3. R script: lab-05-correlation-regression.R. Excel evidence: lab-05-correlation-regression.xlsx. Learner output: Correlation evidence, fitted regression model, diagnostic plot, prediction and limitations statement. Acceptance: The model output is reproducible, coefficient and R-squared interpretations use correct units, diagnostics are reviewed, and the conclusion avoids causal claims.

Lab 6: Test Differences and Compare Groups · Slides 82-83

Alignment: LO3. R script: lab-06-hypothesis-anova.R. Excel evidence: lab-06-hypothesis-anova.xlsx. Learner output: Test-design record, t-test and ANOVA outputs, effect estimates, assumption checks and conclusion. Acceptance: Test choices match the designs, hypotheses were pre-specified, effect estimates accompany p-values, and conclusions state assumptions and practical limitations.

Topic 04: Qualitative Data Analysis · Slides 86-110

Text preparation · tokenisation · stop words · term frequency · word clouds · thematic interpretation

- **Qualitative evidence:** Open-ended responses carry meaning, context and ambiguity that numeric summaries can miss.

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- **Text preparation:** Normalise case and punctuation while preserving the source and recording every exclusion.
- **Tokens:** Break text into reproducible analytical units such as words, phrases or coded themes.
- **Stop words:** Remove high-frequency low-information terms using a documented list and domain judgement.
- **Frequency is not importance:** Common terms reveal salience but require context, concordance and representative excerpts.
- **Credible interpretation:** Use a codebook, peer review and traceable excerpts to manage subjectivity.

Lab 7: Prepare Text and Analyse Term Frequency · Slides 106-107

Alignment: LO4. R script: lab-07-text-analysis.R. Excel evidence: lab-07-text-analysis.xlsx. Learner output: Clean token table, stop-word audit, frequency table, channel comparison and interpretation notes. Acceptance: Every token remains traceable to a source record, exclusions are documented, frequency totals reconcile, and the interpretation uses context rather than frequency alone.

Lab 8: Create and Critique a Word Cloud · Slides 108-109

Alignment: LO4. R script: lab-08-word-cloud.R. Excel evidence: lab-08-word-cloud.xlsx. Learner output: Word-cloud image, ranked-frequency table, sensitivity comparison, excerpt evidence and limitations statement. Acceptance: The visual is reproducible, ranked counts are available, preprocessing sensitivity is documented, and every claim is supported by counts plus traceable excerpts.

Topic 05: Data Visualization · Slides 111-135

Grammar of graphics · quantitative and qualitative plots · customisation · accessibility · findings

- **Question-to-chart fit:** Choose position, length, colour and shape to support the analytical purpose.
- **Grammar of graphics:** Build a plot from data, aesthetics, geoms, statistical transforms, scales, coordinates and facets.
- **Truthful scales:** Units, baselines, aspect ratios and uncertainty must not exaggerate or hide evidence.
- **Accessible design:** Use readable labels, sufficient contrast and redundant encodings beyond colour alone.
- **Layered emphasis:** Guide attention with hierarchy and annotation while keeping the complete context available.
- **Interpret findings:** State the pattern, evidence, uncertainty, decision relevance and limitation.

Lab 9: Build Layered Quantitative Visuals · Slides 131-132

Alignment: LO5. R script: lab-09-ggplot-visuals.R. Excel evidence: lab-09-ggplot-visuals.xlsx. Learner output: A bar chart, faceted distribution plot and annotated scatter plot with a chart-choice rationale. Acceptance: All three plots answer their stated questions, remain legible without colour alone, use truthful scales and carry interpretations supported by the displayed data.

Lab 10: Capstone: Review and Communicate Findings · Slides 133-134

Alignment: LO5. R script: lab-10-capstone.R. Excel evidence: lab-10-capstone.xlsx. Learner output: An auditable analysis bundle with control log, statistical results, three decision visuals and a concise evidence-led recommendation. Acceptance: The bundle reruns from source data, all controls pass or are explained, visuals agree with statistical outputs, and the recommendation distinguishes evidence from judgement.

Resources Required

- R 4.2 or later and RStudio Desktop.
- Learner Guide and ten numbered lab folders.
- One runnable R script and one Excel intake/control/evidence workbook per lab.
- Supplied synthetic CSV data and writable outputs folders.
- Projector for concept briefings, editable native PowerPoint chart examples and demonstrations.
- LMS/TMS access for courseware downloads.

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